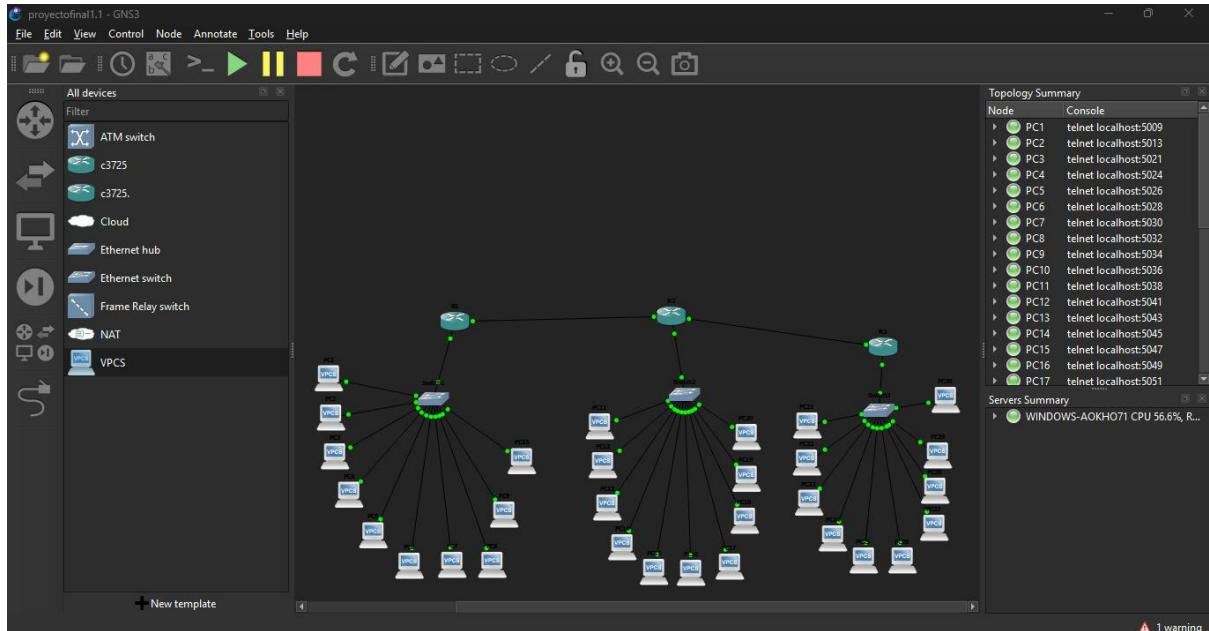


CONFIGURACION EN GNS3

- Topología implementada en gns3



- Estado de las interfaces de los routers mediante el comando Show interface brief.

R1:

```
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.50.145 YES NVRAM    up          up
FastEthernet0/1 unassigned      YES NVRAM    up          up
FastEthernet0/1.20 192.168.50.17 YES NVRAM    up          up
FastEthernet0/1.30 192.168.50.97 YES NVRAM    up          up
FastEthernet1/0 unassigned      YES NVRAM    administratively down down
R1#
```

R2:

```
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.50.146 YES NVRAM    up          up
FastEthernet0/1 192.168.50.149 YES NVRAM    up          up
FastEthernet1/0 unassigned      YES NVRAM    up          up
FastEthernet1/0.10 192.168.50.1 YES NVRAM    up          up
FastEthernet1/0.40 192.168.50.33 YES NVRAM    up          up
FastEthernet1/0.50 192.168.50.49 YES NVRAM    up          up
FastEthernet1/0.60 192.168.50.65 YES NVRAM    up          up
FastEthernet1/0.70 192.168.50.81 YES NVRAM    up          up
FastEthernet1/0.80 192.168.50.105 YES NVRAM    up          up
R2#
```

R3:

```
R3#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.168.50.150 YES NVRAM up up
FastEthernet0/1 unassigned YES NVRAM up up
FastEthernet0/1.90 192.168.50.89 YES NVRAM up up
FastEthernet0/1.100 192.168.50.113 YES NVRAM up up
FastEthernet0/1.110 192.168.50.121 YES NVRAM up up
FastEthernet0/1.120 192.168.50.129 YES NVRAM up up
FastEthernet0/1.130 192.168.50.137 YES NVRAM up up
FastEthernet1/0 unassigned YES NVRAM administratively down down
R3#
```

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- Configuración general de los routers mediante el comando `show running-config`.

Las evidencias muestran la habilitación del protocolo de enlace troncal dot1Q asignado a cada ID de VLAN departamental, seguido por el segmento de configuración de enrutamiento dinámico donde se observa la activación de RIP versión 2.

R1:

```
R1
interface FastEthernet0/0
 ip address 192.168.50.145 255.255.255.252
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface FastEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.50.17 255.255.255.240
!
interface FastEthernet0/1.30
 encapsulation dot1Q 30
 ip address 192.168.50.97 255.255.255.248
!
interface FastEthernet1/0
 no ip address
 shutdown
 duplex auto
 speed auto
!
router rip
 version 2
 network 192.168.50.0
 no auto-summary
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
control-plane
!
```

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R2:

```
interface FastEthernet0/0
ip address 192.168.50.146 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/1
ip address 192.168.50.149 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
!
interface FastEthernet1/0.10
encapsulation dot1Q 10
ip address 192.168.50.1 255.255.255.240
!
interface FastEthernet1/0.40
encapsulation dot1Q 40
ip address 192.168.50.33 255.255.255.240
!
interface FastEthernet1/0.50
encapsulation dot1Q 50
ip address 192.168.50.49 255.255.255.240
!
interface FastEthernet1/0.60
encapsulation dot1Q 60
ip address 192.168.50.65 255.255.255.240
!
interface FastEthernet1/0.70
encapsulation dot1Q 70
ip address 192.168.50.81 255.255.255.248
!
interface FastEthernet1/0.80
encapsulation dot1Q 80
ip address 192.168.50.105 255.255.255.248
!
router rip
version 2
network 192.168.50.0
```

R3:

```
interface FastEthernet0/0
ip address 192.168.50.150 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
!
interface FastEthernet0/1.90
encapsulation dot1Q 90
ip address 192.168.50.89 255.255.255.248
!
interface FastEthernet0/1.100
encapsulation dot1Q 100
ip address 192.168.50.113 255.255.255.248
!
interface FastEthernet0/1.110
encapsulation dot1Q 110
ip address 192.168.50.121 255.255.255.248
!
interface FastEthernet0/1.120
encapsulation dot1Q 120
ip address 192.168.50.129 255.255.255.248
!
interface FastEthernet0/1.130
encapsulation dot1Q 130
ip address 192.168.50.137 255.255.255.248
!
interface FastEthernet1/0
no ip address
shutdown
duplex auto
speed auto
!
router rip
version 2
network 192.168.50.0
no auto-summary
!
ip forward-protocol nd
!
```

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- Pruebas de conectividad entre dispositivos

1. Conectividad en el mismo piso y diferente departamento, en este caso nivel 2

IP PC13: 192.168.50.34

IP PC17: 192.168.50.66

Comandos:

PC13> trace 192.168.50.66

PC13> ping 192.168.50.66

```
PC13> ping 192.168.50.66
192.168.50.66 icmp_seq=1 timeout
84 bytes from 192.168.50.66 icmp_seq=2 ttl=63 time=63.634 ms
84 bytes from 192.168.50.66 icmp_seq=3 ttl=63 time=32.316 ms
84 bytes from 192.168.50.66 icmp_seq=4 ttl=63 time=32.149 ms
84 bytes from 192.168.50.66 icmp_seq=5 ttl=63 time=31.488 ms

PC13> trace 192.168.50.66
trace to 192.168.50.66, 8 hops max, press Ctrl+C to stop
 1  192.168.50.33    16.112 ms  16.656 ms  15.838 ms
 2  *192.168.50.66   31.135 ms (ICMP type:3, code:3, Destination port unreachable)
```

2. Validación de extremo a extremo

IP PC1: 192.168.50.18

IP PC30: 192.168.50.139

Comandos:

PC1> trace 192.168.50.139

PC1> ping 192.168.50.139

```
PC1> trace 192.168.50.139
trace to 192.168.50.139, 8 hops max, press Ctrl+C to stop
 1  192.168.50.17    15.791 ms  16.034 ms  16.564 ms
 2  192.168.50.146   46.832 ms  47.832 ms  46.224 ms
 3  192.168.50.150   78.368 ms  77.451 ms  76.837 ms
 4  **192.168.50.139  92.542 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> ping 192.168.50.139
84 bytes from 192.168.50.139 icmp_seq=1 ttl=61 time=91.613 ms
84 bytes from 192.168.50.139 icmp_seq=2 ttl=61 time=92.177 ms
84 bytes from 192.168.50.139 icmp_seq=3 ttl=61 time=92.286 ms
84 bytes from 192.168.50.139 icmp_seq=4 ttl=61 time=93.275 ms
84 bytes from 192.168.50.139 icmp_seq=5 ttl=61 time=94.184 ms

PC1> █
```